



FATE MONTHLY REPORT

Doc.No : FATE-001
Version : 1.0
Date : 04-10-2023



FORECAST SYSTEM FOR ATMOSPHERE AND TURBULENCE at the VERY LARGE TELESCOPE

Performance assessment
October 2023



PO102958/ESO/20/95952/FLAB

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1 Title for Night 1

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1.1 Title for Figures of Night 1

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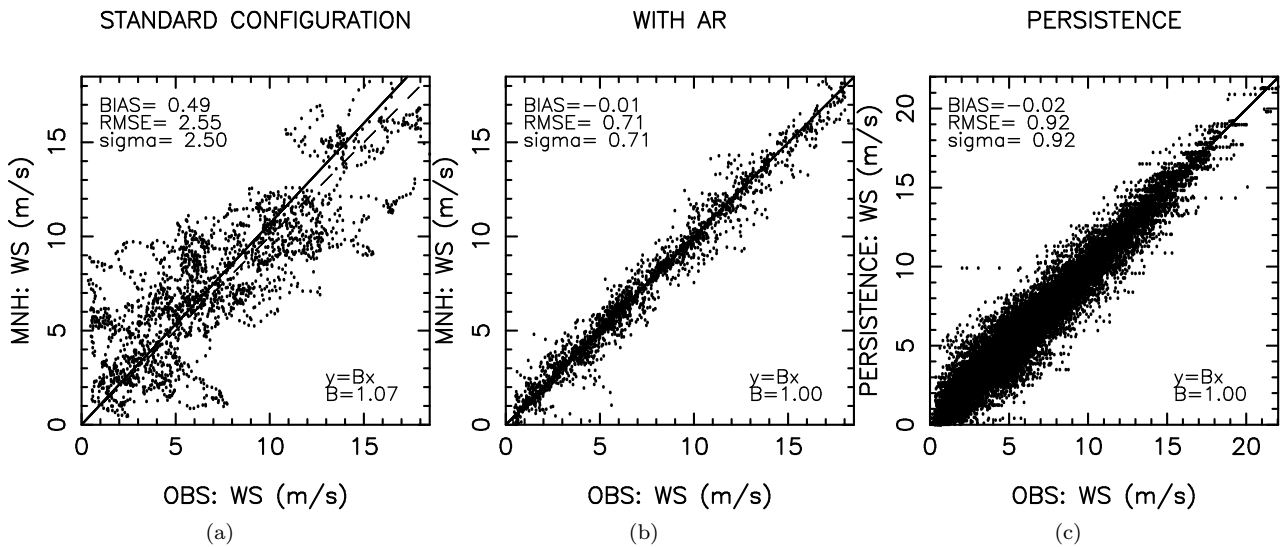


Figure 1: wind speed (ms^{-1}): (a) STANDARD CONFIGURATION, (b) WITH AR (1H), (c) PERSISTENCE (1H).

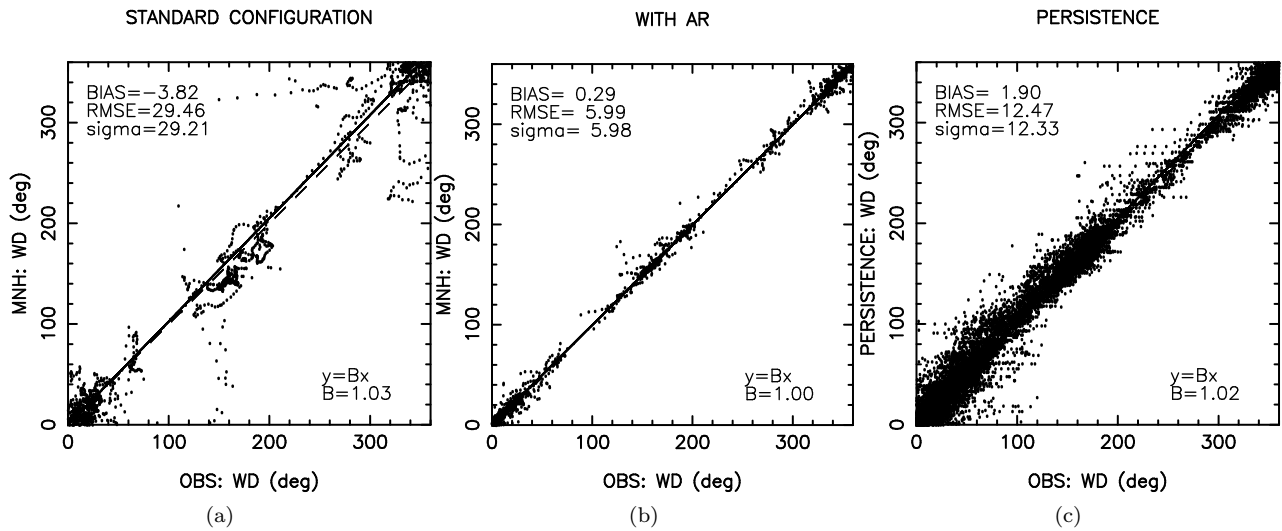


Figure 2: wind direction (degree): (a) STANDARD CONFIGURATION, (b) WITH AR (1H), (c) PERSISTENCE (1H).

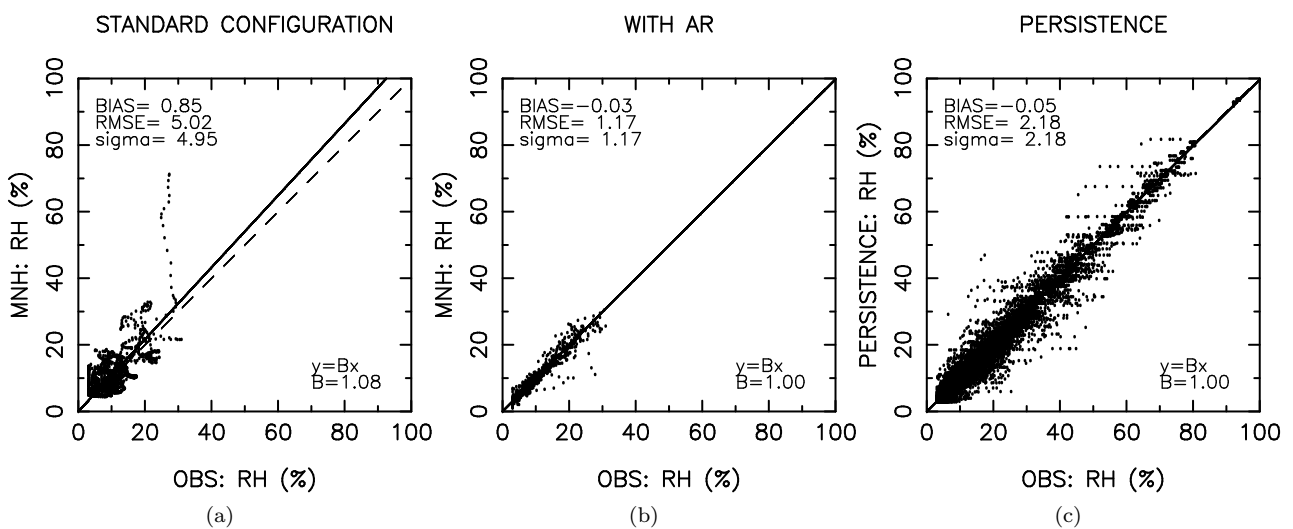


Figure 3: relative humidity (percent): (a) STANDARD CONFIGURATION, (b) WITH AR (1H), (c) PERSISTENCE (1H).

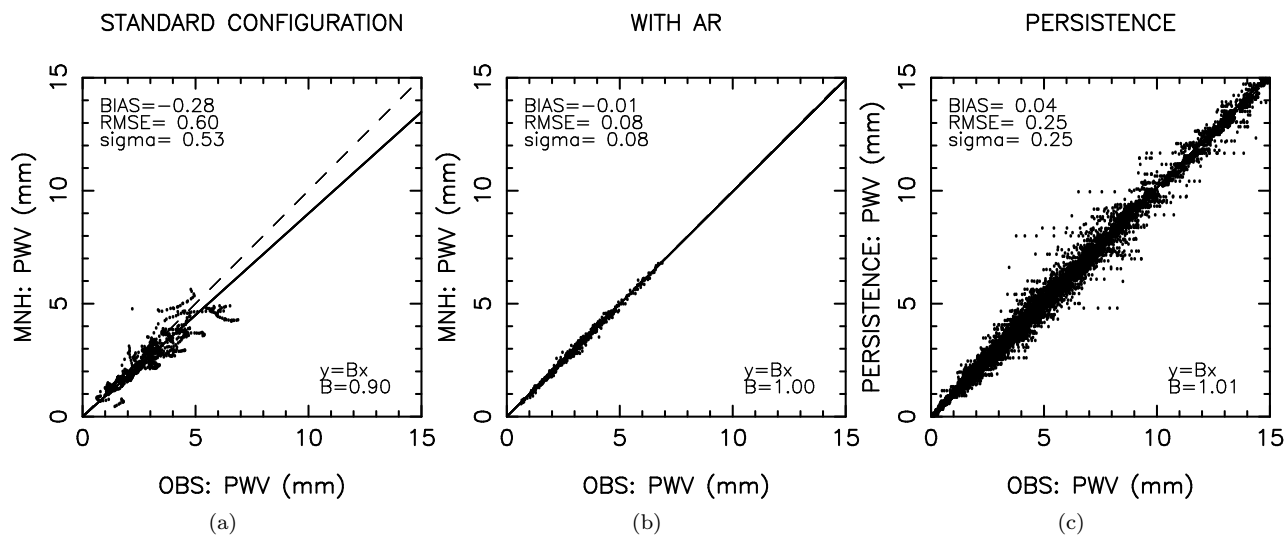


Figure 4: precipitable water vapor (mm): (a) STANDARD CONFIGURATION, (b) WITH AR (1H), (c) PERSISTENCE (1H).

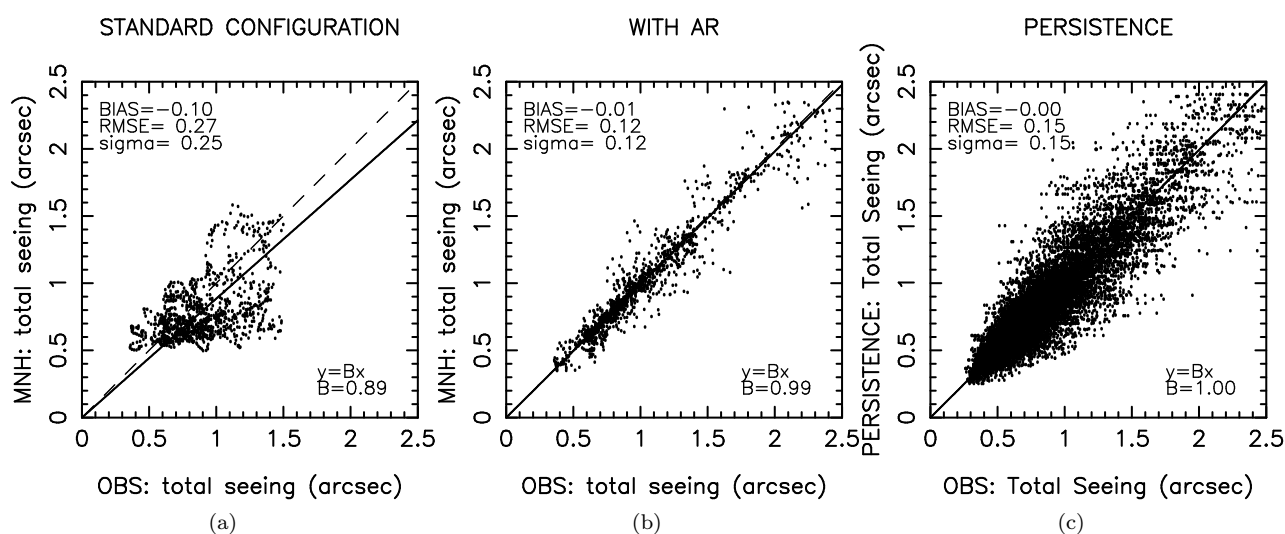


Figure 5: total seeing (arcsec): (a) STANDARD CONFIGURATION ($< 1.5''$), (b) WITH AR (1H), (c) PERSISTENCE (1H).

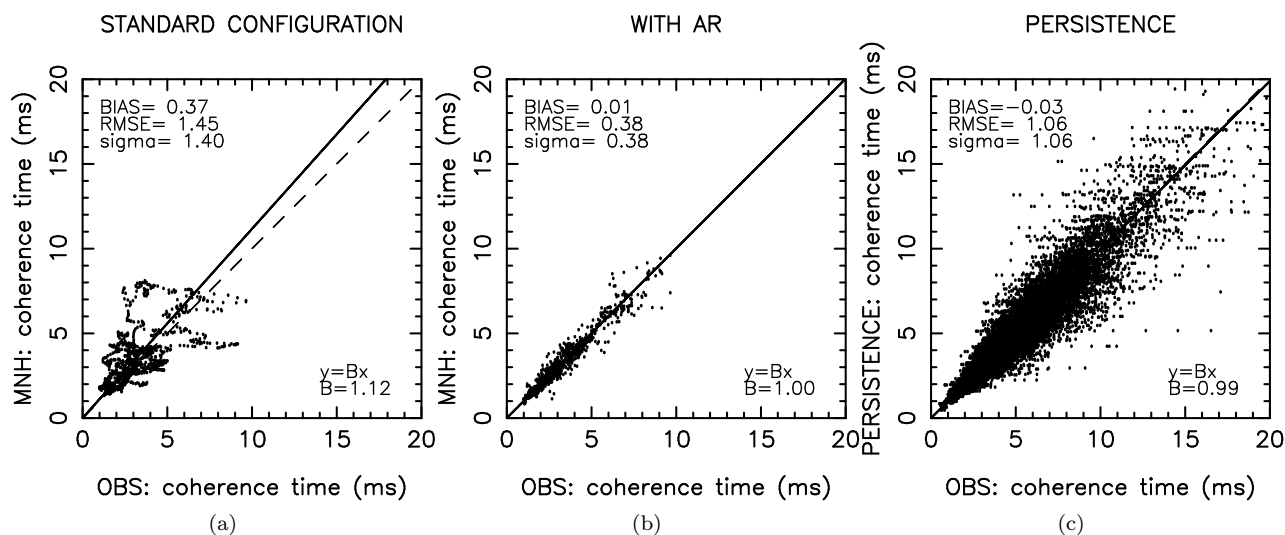


Figure 6: coherence time (ms): (a) STANDARD CONFIGURATION, (b) WITH AR (1H), (c) PERSISTENCE (1H).

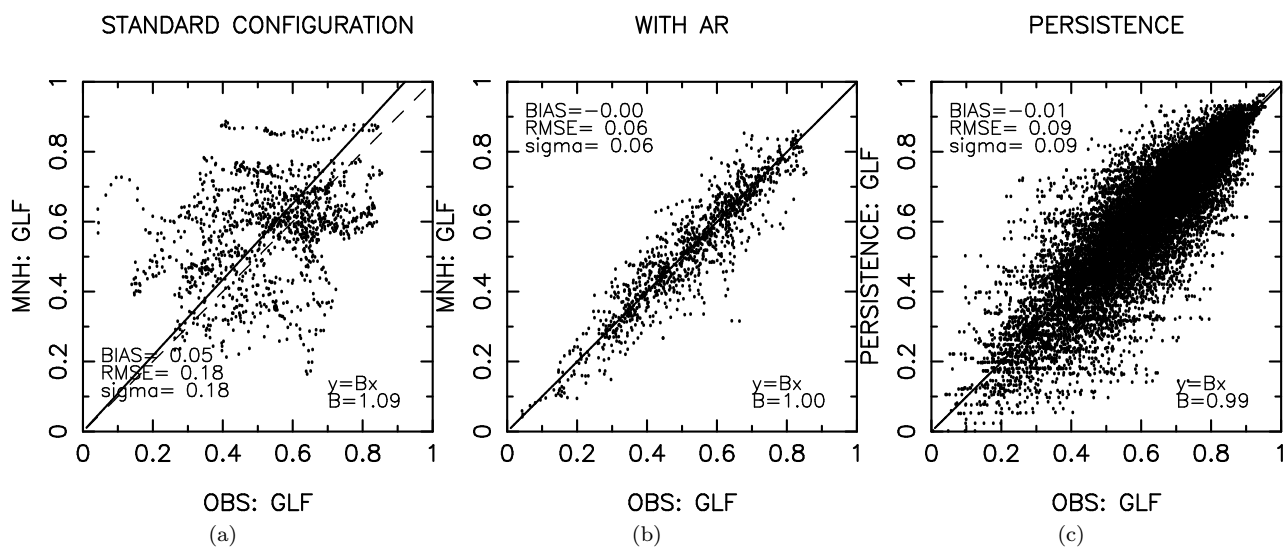


Figure 7: ground layer fraction (dimensionless): (a) STANDARD CONFIGURATION, (b) WITH AR (1H), (c) PERSISTENCE (1H)

1.2 Title for Statistics of Night 1

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VARIABLE	BIAS (inc. month)	RMSE (inc. month)	SD (inc. month)	BIAS (last month)	RMSE (last month)	SD (last month)
WS ($\text{m}\cdot\text{s}^{-1}$)	0.49	2.55	2.50	0.53	2.60	2.55
WD (degree)	-3.82	29.46	29.21	-4.47	30.39	30.06
RH (percent)	0.85	5.02	4.95	0.82	4.94	4.87
PWV (mm)	-0.28	0.60	0.53	-0.28	0.59	0.52
SEE (arcsec)	-0.10	0.27	0.25	-0.10	0.27	0.25
TAU (ms)	0.37	1.45	1.40	0.41	1.51	1.45
GLF (dimensionless)	0.05	0.18	0.18	0.04	0.19	0.18

Table 1: Statistics for variables in standard configuration: incremental month (i.e., since the beginning of service) *vs* last month (September-2023)

FATE - Monthly Report

VARIABLE	BIAS AR (1H)	RMSE AR (1H)	SD AR (1H)		RMSE persistence (1H)	SD persistence (1H)
WS ($m \cdot s^{-1}$)	-0.01	0.72	0.72		0.82	0.82
WD (degree)	0.35	6.18	6.17		10.50	10.46
RH (percent)	-0.03	1.09	1.09		0.91	0.91
PWV (mm)	-0.01	0.08	0.08		0.31	0.31
SEE (arcsec)	-0.01	0.12	0.12		0.17	0.17
TAU (ms)	0.01	0.39	0.39		0.83	0.82
GLF (dimensionless)	-0.00	0.06	0.06		0.09	0.09

Table 2: Last month (September-2023): statistics for variables with AR (1H) *vs* persistence (1H)

VARIABLE	BIAS AR (1H)	RMSE AR (1H)	SD AR (1H)		RMSE persistence (1H)	SD persistence (1H)
WS ($m \cdot s^{-1}$)	-0.01	0.71	0.71		0.92	0.92
WD (degree)	0.29	5.99	5.98		12.47	12.33
RH (percent)	-0.03	1.17	1.17		2.18	2.18
PWV (mm)	-0.01	0.08	0.08		0.25	0.25
SEE (arcsec)	-0.01	0.12	0.12		0.15	0.15
TAU (ms)	0.01	0.38	0.38		1.06	1.06
GLF (dimensionless)	-0.00	0.06	0.06		0.09	0.09

Table 3: Incremental month (i.e., since the beginning of service): statistics for variables with AR (1H) *vs* persistence (1H)

1.3 Title for Contingency tables of Night 1

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wind speed ms^{-1}	Observations		
	ws < 4.7	4.7 < ws < 8.8	ws > 8.8
Model data	ws < 4.7	401	92
	4.7 < ws < 8.8	288	436
	ws > 8.8	25	186
Sample size: 3456; PC=68.0%; EBD=1.2%; POD1=56.2%; POD2=61.1%; POD3=86.7%			

Table 4: Contingency table for wind speed (ms^{-1}) in standard configuration

wind speed ms^{-1}	Observations		
	ws < 4.7	4.7 < ws < 8.8	ws > 8.8
Model data	ws < 4.7	685	37
	4.7 < ws < 8.8	29	650
	ws > 8.8	0	27
Sample size: 3456; PC=94.2%; EBD=0.0%; POD1=95.9%; POD2=91.0%; POD3=95.7%			

Table 5: Contingency table for variable wind speed (ms^{-1}) processed with AR (1H)

FATE - Monthly Report

wind direction degree		North	East	South	Observations	West
Model data	North	1062	31	14		45
	East	22	71	34		0
	South	5	15	345		0
	West	45	2	1		60

Sample size: 3456; PC=87.8%; EBD=1.2%; POD1=93.7%; POD2=59.7%; POD3=87.6%; POD4=57.1%

Table 6: Contingency table for wind direction (degree) in standard configuration

wind direction degree		North	East	South	Observations	West
Model data	North	1125	12	0		3
	East	5	98	3		0
	South	0	9	389		0
	West	4	0	2		102

Sample size: 3456; PC=97.8%; EBD=0.0%; POD1=99.2%; POD2=82.4%; POD3=98.7%; POD4=97.1%

Table 7: Contingency table for wind direction (degree) in standard configuration

FATE - Monthly Report

relative humidity percent		Observations		
		rh < 8.3	8.3 < rh < 11.1	rh > 11.1
Model data	rh < 8.3	422	280	64
	8.3 < rh < 11.1	129	250	163
	rh > 11.1	151	169	473
Sample size: 3456; PC=54.5%; EBD=10.2%; POD1=60.1%; POD2=35.8%; POD3=67.6%				

Table 8: Contingency table for variable relative humidity (percent) in standard configuration

relative humidity percent		Observations		
		rh < 8.3	8.3 < rh < 11.1	rh > 11.1
Model data	rh < 8.3	660	37	2
	8.3 < rh < 11.1	41	625	43
	rh > 11.1	1	37	655
Sample size: 3456; PC=92.3%; EBD=0.1%; POD1=94.0%; POD2=89.4%; POD3=93.6%				

Table 9: Contingency table for variable relative humidity (percent) processed with AR (1H)

precipitable water vapor mm		Observations		
		pwv < 2.1	2.1 < pwv < 3.2	pwv > 3.2
Model data	pwv < 2.1	327	80	0
	2.1 < pwv < 3.2	9	239	100
	pwv > 3.2	0	17	235
Sample size: 1728; PC=79.5%; EBD=0.0%; POD1=97.3%; POD2=71.1%; POD3=70.1%				

Table 10: Contingency table for variable precipitable water vapor (mm) in standard configuration

precipitable water vapor mm		Observations		
		pwv < 2.1	2.1 < pwv < 3.2	pwv > 3.2
Model data	pwv < 2.1	328	11	0
	2.1 < pwv < 3.2	8	316	10
	pwv > 3.2	0	9	325
Sample size: 1728; PC=96.2%; EBD=0.0%; POD1=97.6%; POD2=94.0%; POD3=97.0%				

Table 11: Contingency table for variable precipitable water vapor (mm) processed with AR (1H)

FATE - Monthly Report

total seeing arcsec		Observations		
		see < 0.8	0.8 < see < 1.0	see > 1.0
Model data	see < 0.8	272	219	117
	0.8 < see < 1.0	71	96	118
	see > 1.0	11	39	118
Sample size: 2700; PC=45.8%; EBD=12.1%; POD1=76.8%; POD2=27.1%; POD3=33.4%				

Table 12: Contingency table for variable total seeing (arcsec) in standard configuration and accuracy 0.0

total seeing arcsec		Observations		
		see < 0.8	0.8 < see < 1.0	see > 1.0
Model data	see < 0.8	386	41	1
	0.8 < see < 1.0	28	346	41
	see > 1.0	1	27	372
Sample size: 2700; PC=88.8%; EBD=0.2%; POD1=93.0%; POD2=83.6%; POD3=89.9%				

Table 13: Contingency table for variable total seeing (arcsec) processed with AR (1H) and accuracy 0.0

total seeing arcsec		Observations		
		see < 0.8	0.8 < see < 1.0	see > 1.0
Model data	see < 0.8	347	2	102
	0.8 < see < 1.0	0	345	102
	see > 1.0	7	7	251
Sample size: 2700; PC=88.9%; EBD=10.3%; POD1=98.0%; POD2=97.5%; POD3=55.2%				

Table 14: Contingency table for variable total seeing (arcsec) in standard configuration and accuracy 0.24

total seeing arcsec		Observations		
		see < 0.8	0.8 < see < 1.0	see > 1.0
Model data	see < 0.8	414	0	2
	0.8 < see < 1.0	0	410	2
	see > 1.0	1	4	412
Sample size: 2700; PC=99.4%; EBD=0.2%; POD1=99.8%; POD2=99.0%; POD3=99.0%				

Table 15: Contingency table for variable total seeing (arcsec) processed with AR (1H) and accuracy 0.24

FATE - Monthly Report

coherence time		Observations		
ms		$\tau < 2.3$	$2.3 < \tau < 3.5$	$\tau > 3.5$
Model data	$\tau < 2.3$	238	28	8
	$2.3 < \tau < 3.5$	88	197	95
	$\tau > 3.5$	59	160	282

Sample size: 2700; PC=62.1%; EBD=5.8%; POD1=61.8%; POD2=51.2%; POD3=73.2%

Table 16: Contingency table for variable coherence time (ms) in standard configuration and accuracy 0.0

coherence time		Observations		
ms		$\tau < 2.3$	$2.3 < \tau < 3.5$	$\tau > 3.5$
Model data	$\tau < 2.3$	344	25	0
	$2.3 < \tau < 3.5$	41	336	22
	$\tau > 3.5$	0	24	363

Sample size: 2700; PC=90.3%; EBD=0.0%; POD1=89.4%; POD2=87.3%; POD3=94.3%

Table 17: Contingency table for variable coherence time (ms) processed with AR (1H) and accuracy 0.0

coherence time		Observations		
ms		$\tau < 2.3$	$2.3 < \tau < 3.5$	$\tau > 3.5$
Model data	$\tau < 2.3$	334	0	8
	$2.3 < \tau < 3.5$	0	334	8
	$\tau > 3.5$	51	51	377

Sample size: 2700; PC=90.5%; EBD=5.1%; POD1=86.8%; POD2=86.8%; POD3=95.9%

Table 18: Contingency table for variable coherence time (ms) in standard configuration and accuracy 1.22

coherence time		Observations		
ms		$\tau < 2.3$	$2.3 < \tau < 3.5$	$\tau > 3.5$
Model data	$\tau < 2.3$	385	0	0
	$2.3 < \tau < 3.5$	0	385	0
	$\tau > 3.5$	0	0	385

Sample size: 2700; PC=100.0%; EBD=0.0%; POD1=100.0%; POD2=100.0%; POD3=100.0%

Table 19: Contingency table for variable coherence time (ms) processed with AR (1H) and accuracy 1.22

FATE - Monthly Report

ground layer fraction dimensionless	Observations		
	glf < 0.5	0.5 < glf < 0.6	glf > 0.6
Model data	glf < 0.5	132	74
	0.5 < glf < 0.6	162	143
	glf > 0.6	93	170
Sample size: 2700; PC=39.5%; EBD=11.4%; POD1=34.1%; POD2=37.0%; POD3=47.4%			

Table 20: Contingency table for variable ground layer fraction (dimensionless) in standard configuration and accuracy 0.0

ground layer fraction dimensionless	Observations		
	glf < 0.5	0.5 < glf < 0.6	glf > 0.6
Model data	glf < 0.5	319	53
	0.5 < glf < 0.6	66	275
	glf > 0.6	2	59
Sample size: 2700; PC=79.1%; EBD=0.6%; POD1=82.4%; POD2=71.1%; POD3=83.9%			

Table 21: Contingency table for variable ground layer fraction (dimensionless) processed with AR (1H) and accuracy 0.0

ground layer fraction dimensionless	Observations		
	glf < 0.5	0.5 < glf < 0.6	glf > 0.6
Model data	glf < 0.5	267	13
	0.5 < glf < 0.6	27	350
	glf > 0.6	93	24
Sample size: 2700; PC=82.5%; EBD=12.0%; POD1=69.0%; POD2=90.4%; POD3=81.3%			

Table 22: Contingency table for variable ground layer fraction (dimensionless) in standard configuration and accuracy 0.14

ground layer fraction dimensionless	Observations		
	glf < 0.5	0.5 < glf < 0.6	glf > 0.6
Model data	glf < 0.5	385	0
	0.5 < glf < 0.6	0	385
	glf > 0.6	2	2
Sample size: 2700; PC=99.0%; EBD=0.9%; POD1=99.5%; POD2=99.5%; POD3=95.9%			

Table 23: Contingency table for variable ground layer fraction (dimensionless) processed with AR (1H) and accuracy 0.14

1.4 Title for PODs tables of Night 1

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PARAMETER	STANDARD	WITH AR (1H)
POD1	56.2	95.9
POD2	61.1	91.0
POD3	86.7	95.7
PC	68.0	94.2
EBD	1.2	0.0

Table 24: PODs for wind speed (ms^{-1})

PARAMETER	STANDARD	WITH AR (1H)
POD1	93.7	99.2
POD2	59.7	82.4
POD3	87.6	98.7
POD4	57.1	97.1
PC	87.8	97.8
EBD	1.2	0.0

Table 25: PODs for wind direction (degree)

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PARAMETER	STANDARD	WITH AR (1H)
POD1	60.1	94.0
POD2	35.8	89.4
POD3	67.6	93.6
PC	54.5	92.3
EBD	10.2	0.1

Table 26: PODs for relative humidity (percent)

PARAMETER	STANDARD	WITH AR (1H)
POD1	97.3	97.6
POD2	71.1	94.0
POD3	70.1	97.0
PC	79.5	96.2
EBD	0.0	0.0

Table 27: PODs for precipitable water vapor (mm)

PARAMETER	STANDARD	WITH AR (1H)
POD1	76.8	93.0
POD2	27.1	83.6
POD3	33.4	89.9
PC	45.8	88.8
EBD	12.1	0.2

Table 28: PODs for total seeing (arcsec) and accuray 0.0

PARAMETER	STANDARD	WITH AR (1H)
POD1	98.0	99.8
POD2	97.5	99.0
POD3	55.2	99.0
PC	88.9	99.4
EBD	10.3	0.2

Table 29: PODs for total seeing (arcsec) and accuray 0.24

PARAMETER	STANDARD	WITH AR (1H)
POD1	61.8	89.4
POD2	51.2	87.3
POD3	73.2	94.3
PC	62.1	90.3
EBD	5.8	0.0

Table 30: PODs for coherence time (ms) and accuray 0.0

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PARAMETER	STANDARD	WITH AR (1H)
POD1	86.8	100.0
POD2	86.8	100.0
POD3	95.9	100.0
PC	90.5	100.0
EBD	5.1	0.0

Table 31: PODs for coherence time (ms) and accuray 1.22

PARAMETER	STANDARD	WITH AR (1H)
POD1	34.1	82.4
POD2	37.0	71.1
POD3	47.4	83.9
PC	39.5	79.1
EBD	11.4	0.6

Table 32: PODs for ground layer fraction (dimensionless) and accuray 0.0

PARAMETER	STANDARD	WITH AR (1H)
POD1	69.0	99.5
POD2	90.4	99.5
POD3	81.3	95.9
PC	82.5	99.0
EBD	12.0	0.9

Table 33: PODs for ground layer fraction (dimensionless) and accuray 0.14

2 Title for Day 1

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2.1 Title for Figures of Day 1

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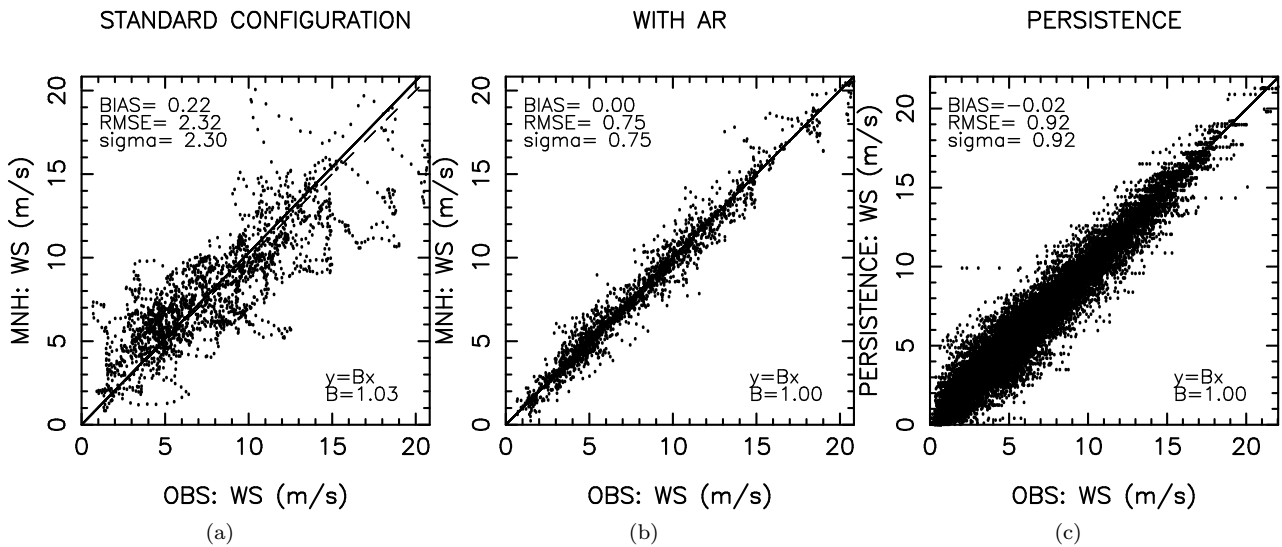


Figure 8: wind speed (ms^{-1}): (a) STANDARD CONFIGURATION, (b) WITH AR (1H), (c) PERSISTENCE (1H).

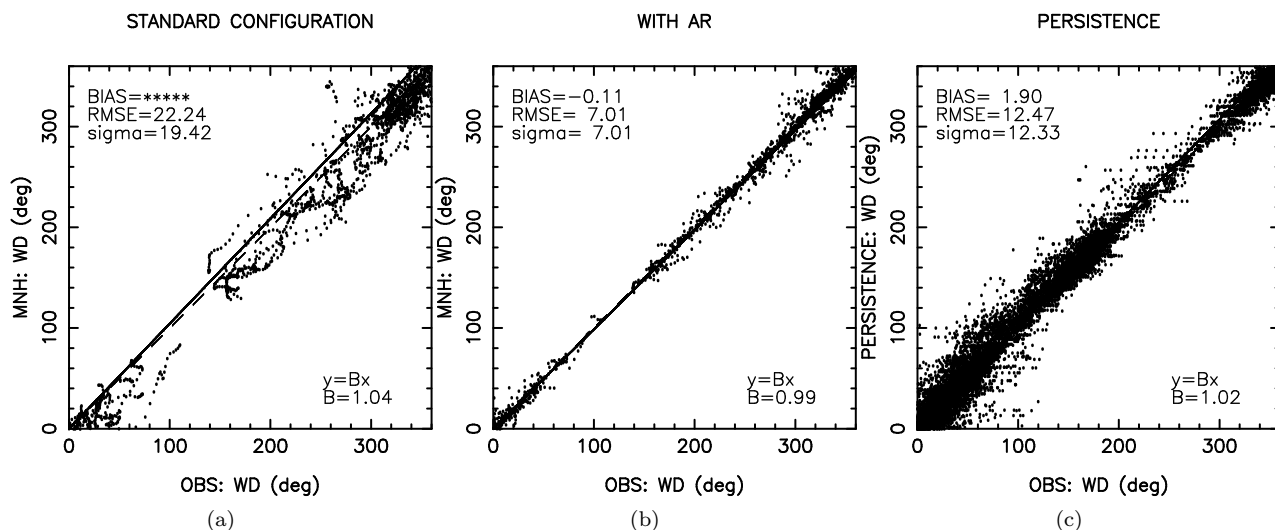


Figure 9: wind direction (degree): (a) STANDARD CONFIGURATION, (b) WITH AR (1H), (c) PERSISTENCE (1H).

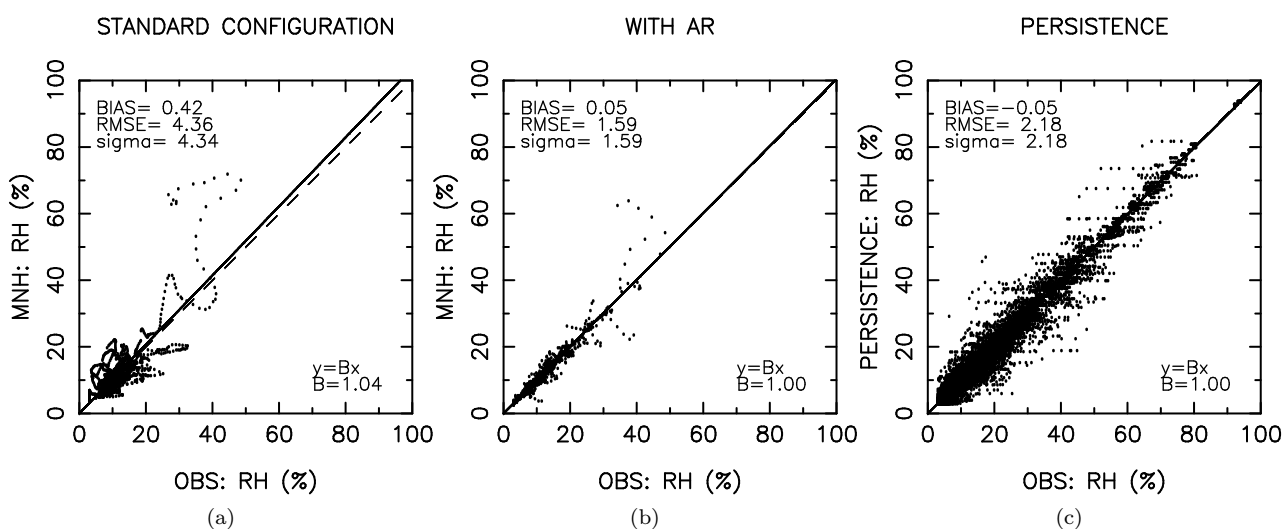


Figure 10: relative humidity (percent): (a) STANDARD CONFIGURATION, (b) WITH AR (1H), (c) PERSISTENCE (1H).

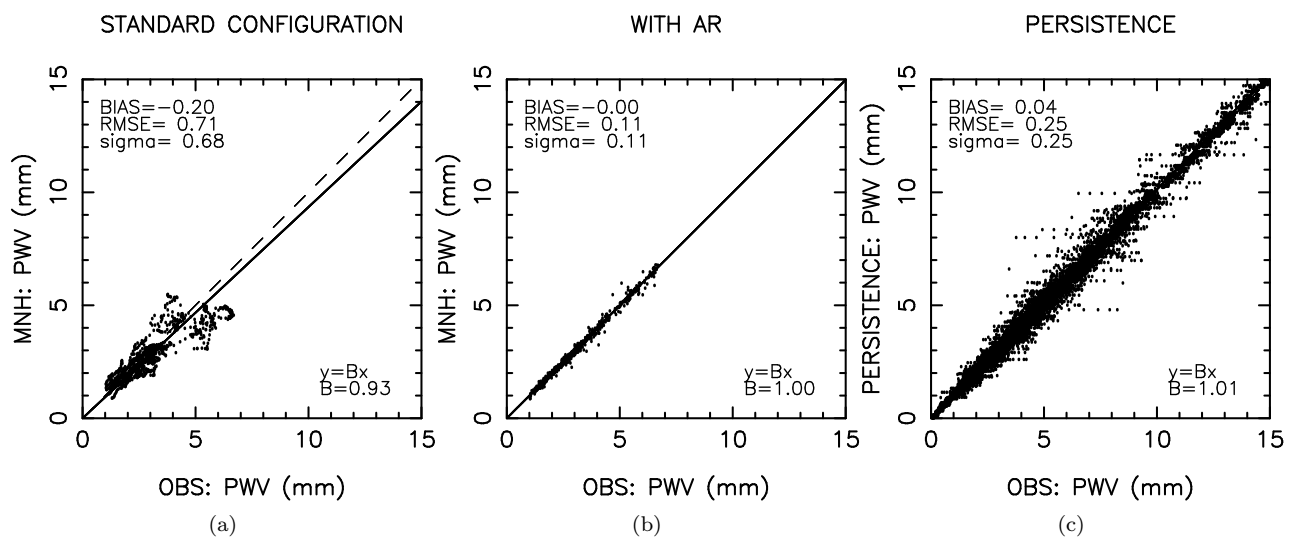


Figure 11: precipitable water vapor (mm): (a) STANDARD CONFIGURATION, (b) WITH AR (1H), (c) PERSISTENCE (1H).

2.2 Title for Statistics of Day 1

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2.3 Title for Contingency tables of Day 1

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wind speed ms^{-1}		Observations		
		ws < 5.2	5.2 < ws < 9.4	ws > 9.4
Model data	ws < 5.2	296	61	0
	5.2 < ws < 9.4	387	536	149
	ws > 9.4	5	91	538
Sample size: 3456; PC=66.4%; EBD=0.2%; POD1=43.0%; POD2=77.9%; POD3=78.3%				

Table 34: Contingency table for wind speed (ms^{-1}) in standard configuration

wind direction degree		Observations			
		North	East	South	West
Model data	North	867	38	0	45
	East	1	42	8	0
	South	0	0	251	73
	West	196	0	10	365
Sample size: 3456; PC=80.4%; EBD=0.0%; POD1=81.5%; POD2=52.5%; POD3=93.3%; POD4=75.6%					

Table 35: Contingency table for wind direction (degree) in standard configuration

relative humidity percent	Observations		
	rh < 9.0	9.0 < rh < 11.6	rh > 11.6
Model data	rh < 9.0	430	240
	9.0 < rh < 11.6	170	251
	rh > 11.6	87	196
Sample size: 3456; PC=58.2%; EBD=5.7%; POD1=62.6%; POD2=36.5%; POD3=75.5%			

Table 36: Contingency table for variable relative humidity (percent) in standard configuration

precipitable water vapor mm	Observations		
	pwv < 2.2	2.2 < pwv < 3.2	pwv > 3.2
Model data	pwv < 2.2	291	71
	2.2 < pwv < 3.2	33	224
	pwv > 3.2	0	29
Sample size: 1728; PC=80.1%; EBD=0.2%; POD1=89.8%; POD2=69.1%; POD3=81.5%			

Table 37: Contingency table for variable precipitable water vapor (mm) in standard configuration

2.4 Title for PODs tables of Day 1

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PARAMETER	STANDARD	WITH AR (1H)
POD1	43.0	89.0
POD2	77.9	86.9
POD3	78.3	92.9
PC	66.4	89.6
EBD	0.2	0.0

Table 38: PODs for wind speed (ms^{-1})

PARAMETER	STANDARD	WITH AR (1H)
POD1	81.5	95.0
POD2	52.5	92.5
POD3	93.3	97.8
POD4	75.6	91.1
PC	80.4	94.3
EBD	0.0	0.0

Table 39: PODs for wind direction (degree)

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PARAMETER	STANDARD	WITH AR (1H)
POD1	62.6	92.4
POD2	36.5	86.5
POD3	75.5	93.9
PC	58.2	90.9
EBD	5.7	0.0

Table 40: PODs for relative humidity (percent)

PARAMETER	STANDARD	WITH AR (1H)
POD1	89.8	99.1
POD2	69.1	96.0
POD3	81.5	96.0
PC	80.1	97.0
EBD	0.2	0.0

Table 41: PODs for precipitable water vapor (mm)